

Azaan Khalfe

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EDUCATION

Arizona State University

Master of Computer Science

Tempe, AZ

Expected 2028

University of Washington

Bachelor of Science in Computer Science

Seattle, WA

June 2024

EXPERIENCE

Software Engineer Intern — Contract

October 2024 – January 2025

Microsoft

Redmond, WA

- Built C#/.NET Azure Function Apps integrating 2 enterprise systems, asynchronously processing Service Bus messages and synchronizing application data with Microsoft Dataverse
- Implemented REST API integrations and 2 layers of production observability using structured logging and Application Insights telemetry to improve monitoring and troubleshooting
- Secured service-to-service communication using Azure Managed Identities and structured application components with dependency injection to improve maintainability and testability
- Developed unit, integration, and end-to-end test suites across 3 testing layers, achieving more than 90% code coverage for core integration workflows

Teaching Assistant

April 2024 – June 2024

University of Washington

Seattle, WA

- Led weekly instruction for 24+ students on browser engineering, web architecture, and software implementation concepts
- Created 8+ technical learning materials and evaluated 200+ assignments for correctness, code quality, and architectural design

PROJECTS

DevEnv MCP Server | *Python, FastMCP, asyncio, Pydantic, psutil, Docker*

- Built a Model Context Protocol server exposing approximately 25 typed tools for Docker orchestration, environment discovery, process management, port inspection, and system health workflows
- Parallelized independent subprocess operations with `asyncio.gather`, reducing virtual-environment discovery latency from 5.1 seconds to 1.2 seconds, a 4.2× speedup
- Designed structured Pydantic response schemas and cross-platform abstractions for Windows, macOS, and Linux system inspection
- Created 80+ unit tests and added confirmation requirements and safety checks for destructive operations

Fantasy Draft Assistant | *React, Chrome Extension, Python, SSE, Sleeper API*

- Built a live fantasy-draft assistant combining 3 data sources: Sleeper draft events, expert rankings, and historical NFL data across a React application and Chrome side panel
- Designed a local synchronization service using server-sent events to keep draft state, roster needs, and recommendations synchronized in real time across 2 user interfaces
- Designed a 2-stage decision pipeline separating player-performance prediction from roster-aware recommendation logic to produce more explainable draft suggestions

NFL Fantasy Prediction Platform | *React, Python, Flask, PostgreSQL, XGBoost*

- Built a multi-source data pipeline for more than 10,000 player-week records using API integrations, rate limiting, and fallback data collection
- Trained position-specific XGBoost models using rolling averages, efficiency metrics, matchup context, and trend features
- Added confidence ranges, reliability indicators, and performance visualizations to make model recommendations easier to evaluate

TECHNICAL SKILLS

Languages: Python, Java, C#, JavaScript, TypeScript, SQL, Go, C++

Frameworks: React, Flask, FastAPI, .NET, Node.js, Pydantic

Cloud/Infrastructure: Azure Functions, Service Bus, Application Insights, Docker, Kubernetes, Git, CI/CD

Databases/Tools: PostgreSQL, TimescaleDB, Redis, Convex

AI/Developer Tools: Model Context Protocol, FastMCP, Claude Code, OpenAI Codex